

Generation Interconnection Request System Impact Study Report

For Queue Project Q17-0812

Ohio Falls 100 kV
November 23, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0812. The Interconnection Customer proposes a 105.5 MW solar photovoltaic facility interconnecting at the Ohio Falls 100 kV substation in the Illinois area. The request was received on June 22, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 314 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 18 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,053,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Sandhill - Crown 100 KV	Elk Grove - Plains 100 KV	129.2	142	21.8
Prospect Hts - Jepsen 100 KV	Prospect Hts - Crown 100 KV Line	124.4	127	14.2
Crown - Prospect Heights 100 KV line	Jepsen - Prospect Heights 100 KV	119.8	139	15.7
Stapley - Arlington Hts 100 KV	Sandhill - Crown 100 KV Line	112.2	131	11.5
Waldo - Truman 230 KV line	Trego Substation - Beaver 500 KV	111.4	546	9.9
Plains - Elk Grove 100 KV line	Estill Springs 100/1 kV TX	109.8	300	19.6
Waldo Substation 500-230 kV Transformer Bank	Trego - Beaver 500 KV	107.0	716	9.8
E Ridge Substation - Combined Locks 100 KV line	Combined Locks - Wommack 100 KV	106.1	142	20.6
Minnehaha - Sta S Van Nuys 230 KV line	Troy Energy Llc - Richland 500 KV	105.8	786	17.1
Prospect Hts - Potter County 100 KV	Plains - Elkgrove 100 KV Line	105.7	263	38.1
Elk Grove - Davey Town 100 KV Line	Prospect Heights - Potter County 100 KV	105.3	180	21.0
Hebo - Burris Pit 230 KV line	Trego - Beaver 500 KV	104.4	777	8.0
Burris Pit - James De Young 230 KV	Trego - Beaver 500 KV L	103.6	801	8.0
Crown - Jepsen 100 KV	Prospect Heights - Crown 100 KV	103.2	110	14.2
Herald-Twin Cities - Mason St 230 KV line	Mason St - New Departure 230 KV L	103.1	610	5.5
Truman - Lookingglass Substation 230 KV	Beaver Substation - Trego 500 KV	101.8	575	9.9

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
E - Potter County 100 KV	Dell - Potter County 100 KV	101.3	201	27.3
Sta S Van Nuys - Gould 230 KV	Troy Energy Llc - Richland 500 KV	100.6	511	6.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sandhill - Crown 100 KV	Rebuild line with higher-capacity conductor	\$1,549,000	17% (\$270,000)
Prospect Hts - Jepsen 100 KV	Rebuild line with higher-capacity conductor	\$639,000	20% (\$129,000)
Prospect Hts - Crown 100 KV line	Rebuild line with higher-capacity conductor	\$1,709,000	20% (\$346,000)
Arlington Hts - Stapey 100 KV	Rebuild line with higher-capacity conductor	\$776,000	19% (\$144,000)
Waldo - Truman 230 KV	Rebuild line with higher-capacity conductor	\$20,966,000	1% (\$229,000)
Plains - Elk Grove 100 KV line	Reconductor line	\$1,014,000	17% (\$170,000)
Waldo 500-230 kV XFMR	Replace transformer with higher-rated unit	\$34,262,000	1% (\$375,000)
Combined Locks - East Ridge 100 KV line	Reconductor line	\$1,520,000	16% (\$241,000)
Minnehaha - Sta S Van Nuys 230 KV line	Reconductor line	\$7,140,000	1% (\$79,000)
Prospect Hts - Potter County 100 KV line	Reconductor line	\$620,000	22% (\$134,000)
Davey Town - Elk Grove 100 KV	Reconductor line	\$958,000	17% (\$160,000)
Hebo - Burris Pit 230 KV Line	Reconductor line	\$5,284,000	1% (\$57,000)
Burris Pit - James De Young 230 KV	Reconductor line	\$18,610,000	1% (\$202,000)
Crown - Jepsen 100 KV	Reconductor line	\$419,000	20% (\$85,000)
Herald-Twin Cities - Mason St 230 KV line	Reconductor line	\$15,538,000	1% (\$171,000)
Truman - Lookingglass Substation 230 KV	Reconductor line	\$8,136,000	1% (\$89,000)
East - Potter County 100 KV	Reconductor line	\$503,000	23% (\$116,000)
Station S Van Nuys - Gould Substation 230 KV	Reconductor line	\$5,106,000	1% (\$56,000)
	Total allocated to project		\$3,053,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.